

CSE260 (QUIZ - 1 (33))

Summer2025

Fullmark: 15

ID: _____

Name: _____

Section: _____

Q1: Convert this number from Octal to Hexadecimal : $(5463.324)_8$ [4]

$$\begin{array}{ccccccc}
 5 & 4 & 6 & 3 & . & 3 & 2 & 4 \\
 (101100110011.011010100000)_2 \\
 (B33.6A0)_{16}
 \end{array}$$

Q2 : Do the following 7-base multiplication : $(3456)_7 \times (56)_7$ [5]

$$\begin{array}{r}
 \begin{array}{cccc}
 & 4 & 5 & 5 \\
 3 & 4 & 5 & 6 \\
 \times & 5 & 6 \\
 \hline
 3 & 1 & 1 & 0 & 1 \\
 2 & 4 & 3 & 1 & 2 & \times \\
 \hline
 (304221)_7
 \end{array}
 \end{array}$$

$[0-6]$
 $6 \times 6 = 36$
 $7 \overline{)36} (5 \rightarrow \text{carry}$
 $\underline{35}$
 $\quad 1$

Q3. Do the following two maths using 2's complement and identify if there is any carry or overflow. If yes, mention and identify them.

[6]

a) $30 - 15$ [6 bit registers]

b) $-2 - 7$ [4 bit registers]

$$\begin{array}{r}
 \text{a) } +30 : \quad 011110 \\
 -15 : \quad 110001 \\
 \hline
 +15 : \quad 000111 \\
 \text{Carry}
 \end{array}$$

$$\begin{array}{r}
 15 : 001111 \\
 1's : 110000 \\
 +1 \\
 \hline
 -15 : 110001
 \end{array}$$

$$\begin{array}{r}
 \text{b) } +2 : 0010 \\
 \quad \quad 1101 \\
 \quad \quad +1 \\
 \hline
 -2 : 1110
 \end{array}$$

$$\begin{array}{r}
 +7 : 0111 \\
 \quad \quad 1000 \\
 \quad \quad +1 \\
 \hline
 -7 : 1001
 \end{array}$$

$$\begin{array}{r}
 -2 : 1110 \\
 -7 : 1001 \\
 \hline
 1011 \\
 \text{Carry} \quad \text{overflow}
 \end{array}$$

we are adding 2 neg numbers
but getting + ans.